

FACULTY OF COMPUTING UNIVERSITI TEKNOLOGI MALAYSIA

DATA MINING (SECP2753)
SEM II – 2024/2025

**PROJECT 2 – UNSUPERVISED LEARNING
(CLUSTERING)**

INSTRUCTIONS (SECTION A): Group Task

1. Use the same dataset from Project 1. Alternatively, you may select a different dataset if it provides new business insights.
2. Select **TWO (2)** clustering algorithms to perform unsupervised learning.
3. Apply appropriate data processing techniques, such as handling missing values, normalization and feature selection.
4. Build and evaluate clustering models to derive 2-3 meaningful business insights.
5. Visualize the clustering results using suitable charts and graphs.
6. Clearly explain your data pre-processing methods and analysis results in the project report.
7. Submit a group report of 10-20 pages, including figures, tables and relevant outputs.
8. Save and submit your report as: **groupname_project2name.pdf**.

Note : You are allowed to use a different dataset if you aim to derive alternative business insights compared to Project 1.

INSTRUCTIONS (SECTION B): Individual Task (Project Reflection)

This section consists of two parts to be completed individually:

a) **PART A: PROJECT REFLECTION**

- Provide a summary of the tasks conducted in Project 1 and Project 2.
- Offer critical comments on the processes and challenges faced.
- Reflect on the knowledge and skills gained during the project execution.

b) **PART B: COURSE REFLECTION**

- Share your overall reflection on the course.
- Evaluate whether you have met the intended learning outcomes. If not, explain why.
- Discuss the new skills you have acquired from this course.
- Explain how you plan to apply what you've learned in future academic or professional contexts.

ASSESSMENT CRITERIA (20%):

☐ Marks will be awarded based on the following criteria:

❖ SECTION A (15%)

- Explanation of data processing and sampling procedures (3%).
- Construction of clustering models and result visualization (5%).
- Interpretation and discussion of results (5%).
- Quality of technical report (2%).

❖ SECTION B (5%)

- Critical summary of Project 1 and Project 2 tasks.
- Reflection on course outcomes, personal learning and skills acquisition.
- Relevance and depth of the personal reflection.

☐ Submit your complete Project 2 report via e-learning by **4th July 2025 (before 10PM)**.